Avex schedule leveling

Assumptions.

All data presented is based of averages of standard inspections.

Work demand does not include any (drop-ins, heavy structural repair, engine changes, or avionics installations.

All labor estimates are using simple models and do not include things like, variance in efficiency, training burdens, or PTO.

These statistics should inform the greater conversation but not be used for detail planning.

Forecasted inspections / Original data

Location	HRS	Tail #	Next Annual	Annual Due
BJC	171	N930GW	A+ Annual	01/2023
BJC	309	N7SA	C+ Annual	01/2023
BJC	255	N46D	C+ Annual	01/2023
BJC	154	N940CX	A Inspection	02/2023
BJC	201	N850KD	A+ Annual	02/2023
BJC	171	N188CD	A+ Annual	02/2023
BJC	225	N408GT	B+ Annual	02/2023
BJC	256	N850KK	C+ Annual	02/2023
BJC	256	N850HA	C+ Annual	02/2023
BJC	171	N900TK	A+ Annual	03/2023
BJC	188	N977MD	B+ Annual	03/2023
BJC	188	N900AZ	B+ Annual	03/2023
BJC	188	N893MA	B+ Annual	03/2023
BJC	188	N930DX	B+ Annual	03/2023
BJC	255	N910LM	C+ Annual	03/2023
BJC	256	N850WW	C+ Annual	03/2023
BJC	309	N428Z	C+ Annual	03/2023
BJC	255	N850AB	C+ Annual	03/2023
BJC	171	N940HW	A+ Annual	04/2023
BJC	206	N700KH	A+ Annual	04/2023
BJC	171	N940TM	A+ Annual	04/2023
BJC	188	N910GE	B+ Annual	04/2023
BJC	255	N900ZP	C+ Annual	04/2023

CMA and BJC work and capacity forecast 2023

Problems

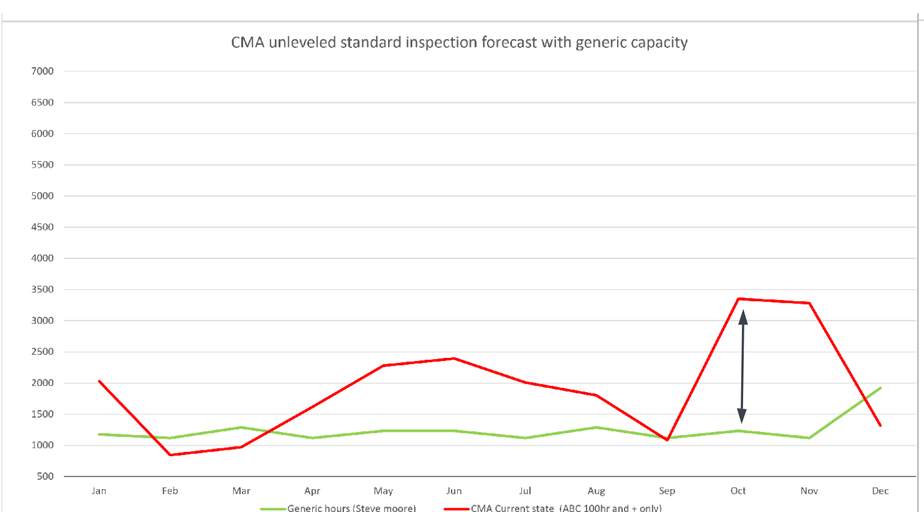
Overcapacity

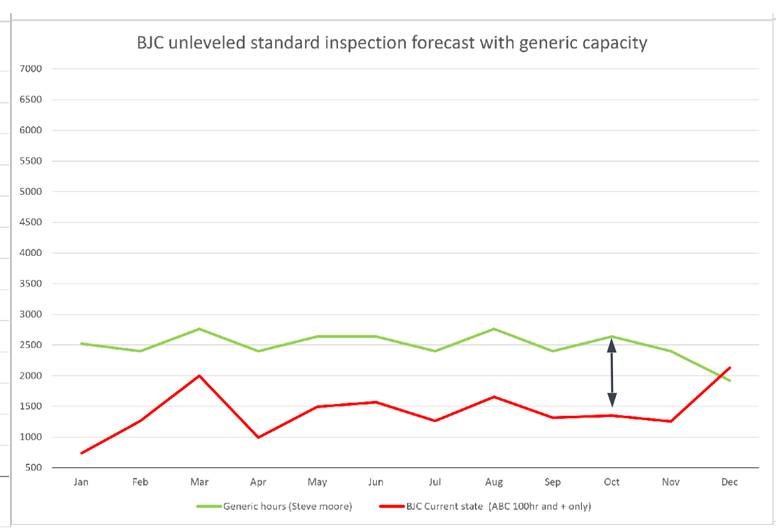
Under capacity

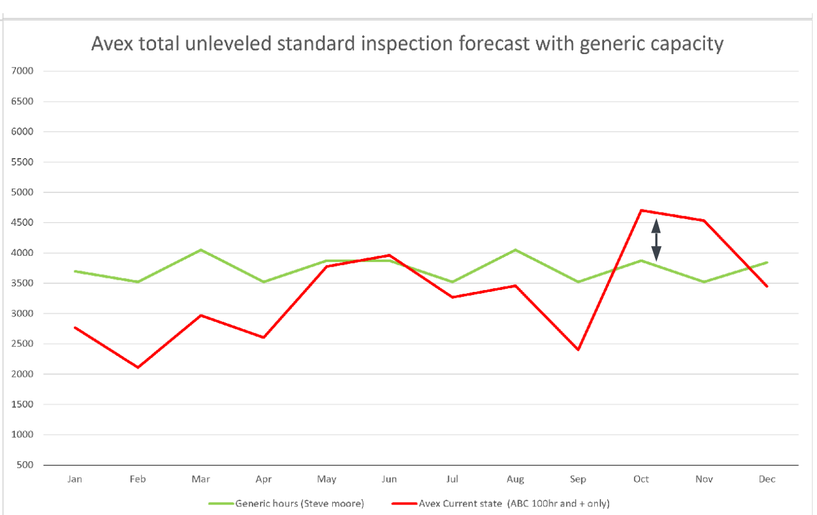
Demand is not smooth

Work Type	Jan CMA	Jan BJC	Feb CMA	Feb BJC	Mar CMA	Mar BJC	Apr CMA	Apr BJC	May CMA	May BJC	Jun CMA	Jun BJC	Jul CMA	Jul BJC	Aug CMA	Aug BJC	Sep CMA	Sep BJC	Oct CMA	Oct BJC	Nov CMA	Nov BJC	Dec CMA	Dec BJC	
100Hr insp hours	80	80	80	0	0	0	80	0	80	0	80	0	80	0	80	0	80	0	80	0	80	0	0	0	
100hr ACFT #	1	1	1	0	0	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	0	0	
A total hours	579	193	579	772	0	0	772	579	1351	1544	965	772	772	579	1158	772	386	772	1351	1158	1737	772	386	772	
A ACFT #	3	1	3	4	0	0	4	3	7	8	5	4	4	3	6	4	2	4	7	6	9	4	2	4	
B total hours	211	0	211	422	844	633	211	422	633	422	422	633	422	211	211	422	422	0	633	0	1055	0	422	211	
B ACFT #	1	0	1	2	4	3	1	2	3	2	2	3	2	1	1	2	2	0	3	0	5	0	2	1	
C total hours	819	1911	0	546	0	819	546	273	0	0	1092	273	819	546	546	546	273	546	1365	273	819	1092	546	819	
C ACFT #	3	7	0	2	0	3	2	1	0	0	4	1	3	2	2	2	1	2	5	1	3	4	2	3	
Total hours	**1689**	**2184**	**870**	**1740**	**844**	**1452**	**1609**	**1274**	**2064**	**1966**	**2559**	**1678**	**2093**	**1336**	**1995**	**1740**	**1161**	**1318**	**3429**	**1431**	**3691**	**1864**	**1354**	**1802**	43143
Total Aircraft	**8**	**9**	**5**	**8**	**4**	**6**	**8**	**6**	**11**	**10**	**12**	**8**	**10**	**6**	**10**	**8**	**6**	**6**	**16**	**7**	**18**	**8**	**6**	**8**	204
Days in month	21	21	20	20	23	23	20	20	22	22	22	22	20	20	23	23	20	20	22	22	20	20	16	16	
8	168	168	160	160	184	184	160	160	176	176	176	176	160	160	184	184	160	160	176	176	160	160	128	128	
Total hours	1176	2520	1120	2400	1288	2760	1120	2400	1232	2640	1232	2640	1120	2400	1288	2760	1120	2400	1232	2640	1120	2400	896	1920	
Difference	**-513**	**336**	**250**	**660**	**444**	**1308**	**-489**	**1126**	**-832**	**674**	**-1327**	**962**	**-973**	**1064**	**-707**	**1020**	**-41**	**1082**	**-2197**	**1209**	**-2571**	**536**	**-458**	**118**	

October Imbalance







Explanation

Base Changes

Schedule changes

Final base location 2023

Location	HRS	Tail #	Original	Scheduled		Next	Original	Annual	Month change
BJC	193	N701AV		Yes		A	01/2023	02/2023	+1
CMA	193	N945DM				A	01/2023	02/2023	+1
BJC	80	N960F	CMA			100	02/2023	01/2023	-1

Original/ traditional base location

New base for leveling

Original forecast for time inspection is due

Corrected time for leveling

Projected aircraft standard inspections and their changes in base and schedule.

Location	HRS	Tail #	Original	Scheduled		Next	Original	Annual	Month change
BJC	193	N701AV		Yes		A	01/2023	02/2023	+1
CMA	193	N945DM				A	01/2023	02/2023	+1
BJC	80	N960F	CMA			100	02/2023	01/2023	-1
BJC	193	N940CX				A	02/2023	01/2023	-1
BJC	193	N850KD		Yes	CONFLICT	A	02/2023	01/2023	-1
BJC	193	N188CD		Yes	CONFLICT	A	02/2023	01/2023	-1
BJC	193	N900TK		Yes	CONFLICT	A	02/2023	01/2023	-1
BJC	273	N850HA				C	02/2023	01/2023	-1
CMA	193	N700DZ				A	02/2023	03/2023	-1
BJC	193	N850AH	CMA			A	04/2023	02/2023	-2
BJC	211	N850LL	CMA			B	04/2023	02/2023	-2
CMA	273	N700AN				C	04/2023	02/2023	-2
CMA	273	N850RB				C	04/2023	02/2023	-2
BJC	193	N525WG	CMA			A	05/2023	03/2023	-1
BJC	193	N977CA	CMA			A	05/2023	04/2023	-1
BJC	193	N43BV	CMA			A	05/2023	04/2023	-1
CMA	193	N31MC				A	05/2023	03/2023	-2
CMA	193	N930VL				A	05/2023	03/2023	-2
CMA	211	N130TM				B	05/2023	04/2023	-1
BJC	193	N668W				A	05/2023	04/2023	-1
BJC	193	N900TY	CMA			A	06/2023	03/2023	-3
BJC	193	N212BW	CMA			A	06/2023	03/2023	-3
BJC	193	N854UC	CMA			A	06/2023	03/2023	-3

Total changes needed for leveling.

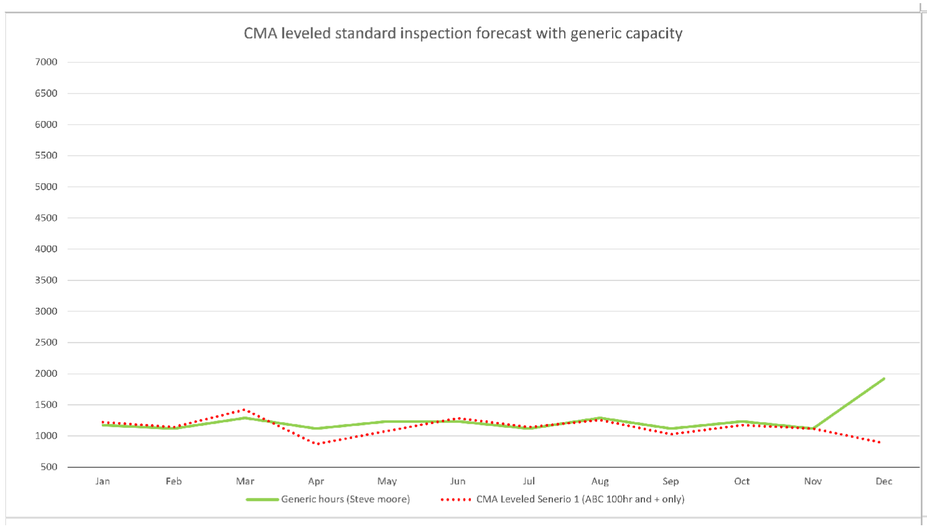
*this is an initial estimate, the scenario can be run many different ways.

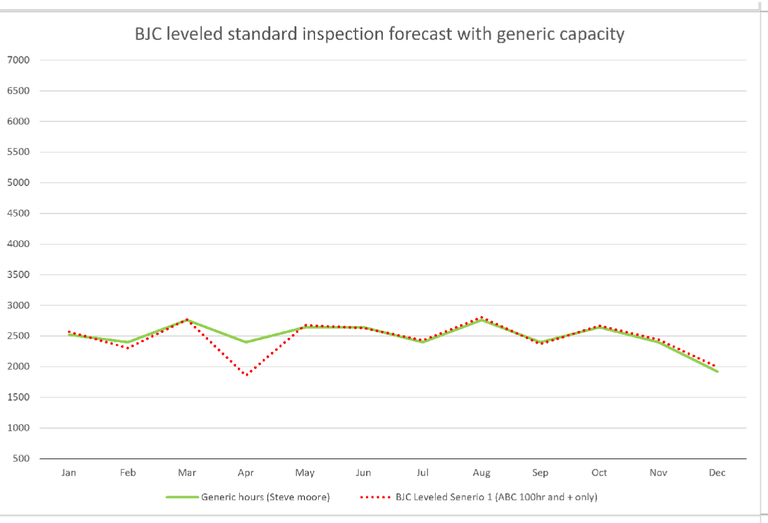
Base	Month	
48	67	
Months	Number	Percentage of total
+1	2	3%
-1	37	55%
-2	22	33%
-3	4	6%
-4	2	3%
-5	0	0%

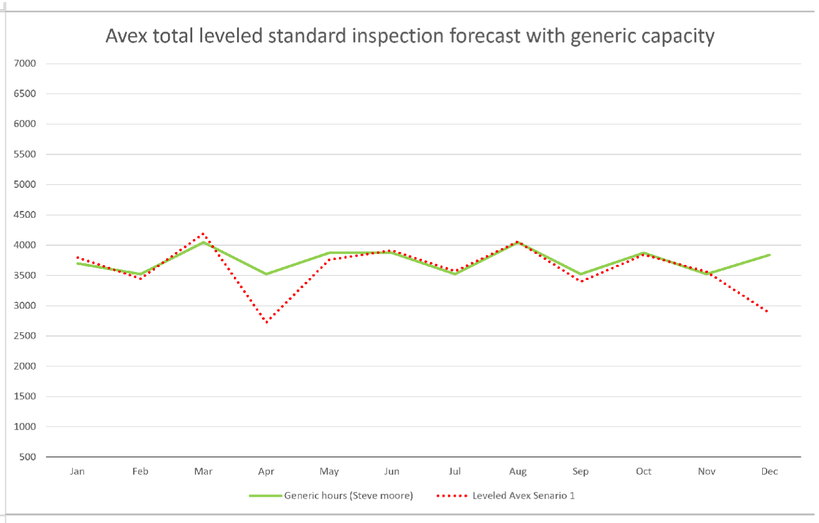
This results in:

Work Type	Jan CMA	Jan BJC	Feb CMA	Feb BJC	Mar CMA	Mar BJC	Apr CMA	Apr BJC	May CMA	May BJC	Jun CMA	Jun BJC	Jul CMA	Jul BJC	Aug CMA	Aug BJC	Sep	Sep	Oct	Oct BJC	Nov CMA	Nov	Dec	Dec
100Hr insp hours	80	160	0	0	0	0	80	0	80	0	80	0	80	80	80	80	0	0	0	0	80	0	0	0
A total hours	386	772	386	579	579	772	579	1158	579	1351	386	965	579	772	965	1544	0	579	965	2123	193	1351	193	965
B total hours	211	0	211	633	844	633	211	422	422	1055	0	844	211	211	211	633	211	422	211	0	844	0	422	211
C total hours	546	1638	546	1092	0	1365	0	273	0	273	819	819	273	1365	0	546	819	1365	0	546	0	1092	273	819
100hr ACFT #	1	2	0	0	0	0	1	0	1	0	1	0	1	1	1	1	0	0	0	0	1	0	0	0
A ACFT #	2	4	2	3	3	4	3	6	3	7	2	5	3	4	5	8	0	3	5	11	1	7	1	5
B ACFT #	1	0	1	3	4	3	1	2	2	5	0	4	1	1	1	3	1	2	1	0	4	0	2	1
C ACFT #	2	6	2	4	0	5	0	1	0	1	3	3	1	5	0	2	3	5	0	2	0	4	1	3
Total hours	**1223**	**2570**	**1143**	**2304**	**1423**	**2770**	**870**	**1853**	**1081**	**2679**	**1285**	**2628**	**1143**	**2428**	**1256**	**2803**	**1030**	**2366**	**1176**	**2669**	**1117**	**2443**	**888**	**1995**
Total Aircraft	6	12	5	10	7	12	5	9	6	13	6	12	6	11	7	14	4	10	6	13	6	11	4	9
Days in month	21	21	20	20	23	23	20	20	22	22	22	22	20	20	23	23	20	20	22	22	20	20	16	16
8	168	168	160	160	184	184	160	160	176	176	176	176	160	160	184	184	160	160	176	176	160	160	128	128
Total hours	1176	2520	1120	2400	1288	2760	1120	2400	1232	2640	1232	2640	1120	2400	1288	2760	1120	2400	1232	2640	1120	2400	896	1920
Difference	-47	-50	-23	96	-135	-10	250	547	151	-39	-53	12	-23	-28	32	-43	90	34	56	-29	3	-43	8	-75

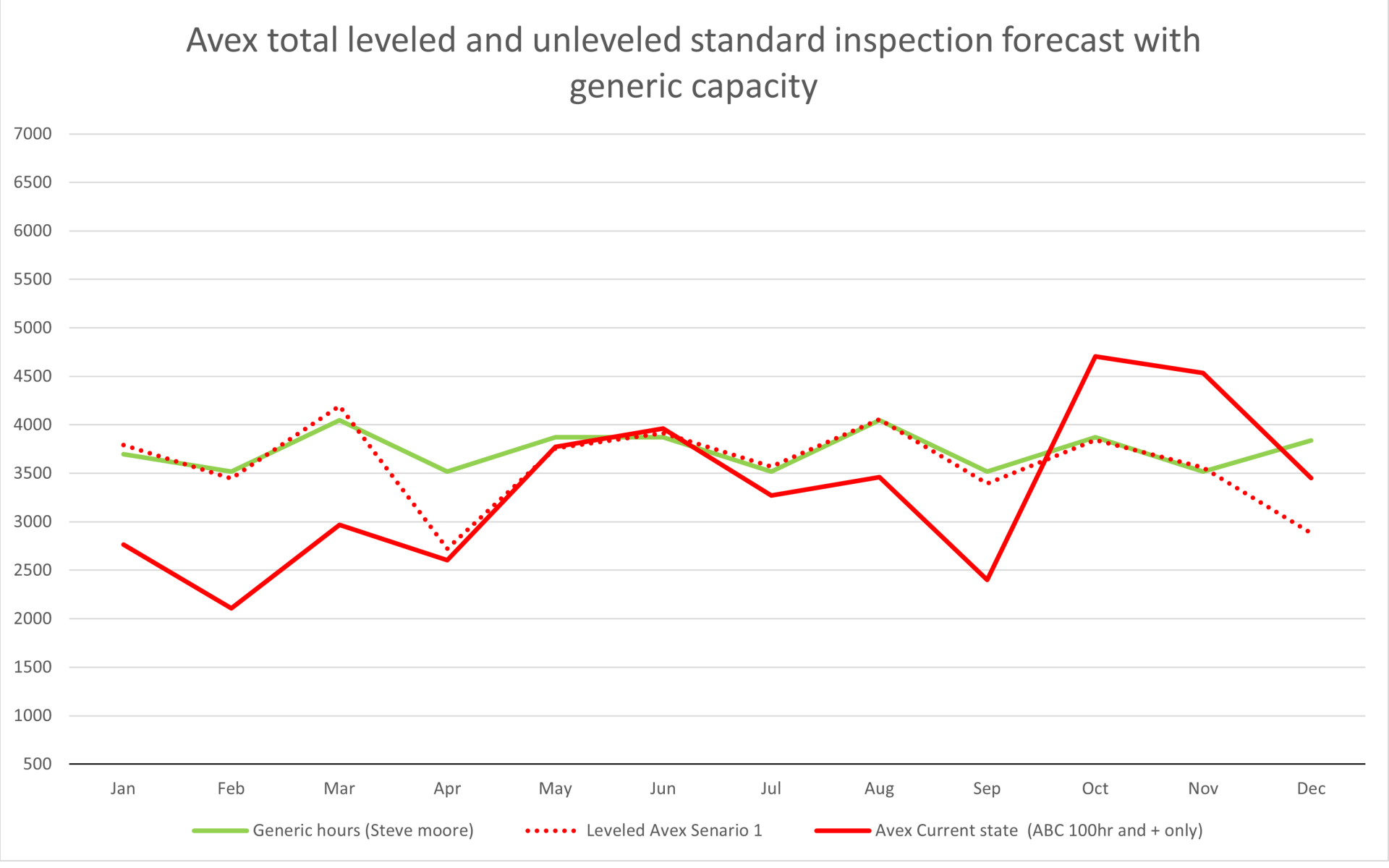
Balanced demand compared to total available hours between both CMA and BJC







Before and after leveling



Next steps:

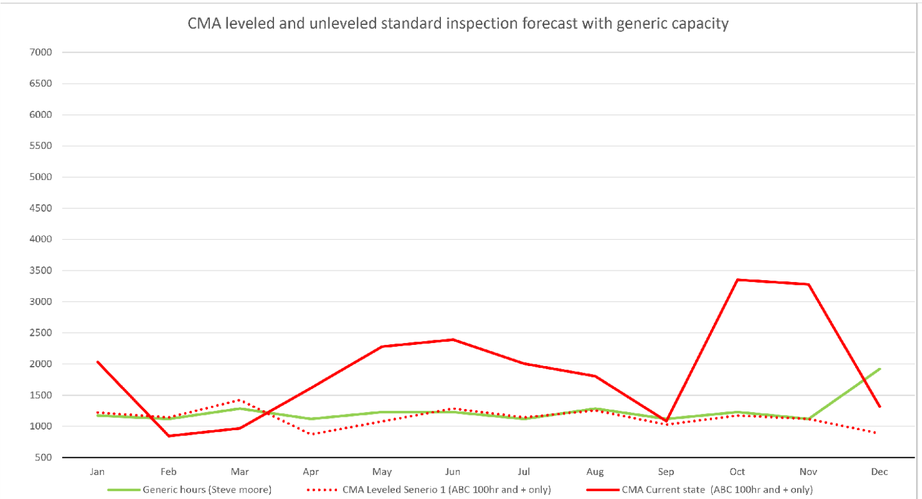
Create list of customers who will collaborate with Avex to move their schedule.

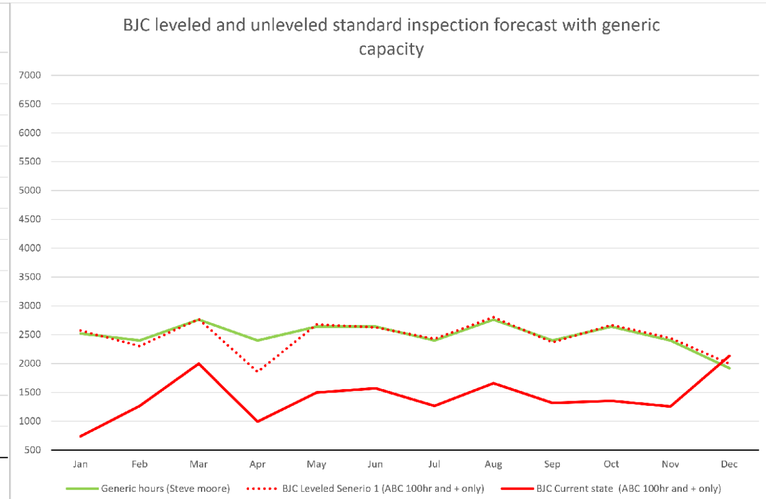
· High usage customers who are hour driven rather by calendar constraints.
· Easy to work with customers.
· If there are not enough easy targets then determine a way to pole the Avex customer base to look for additional collaborators.

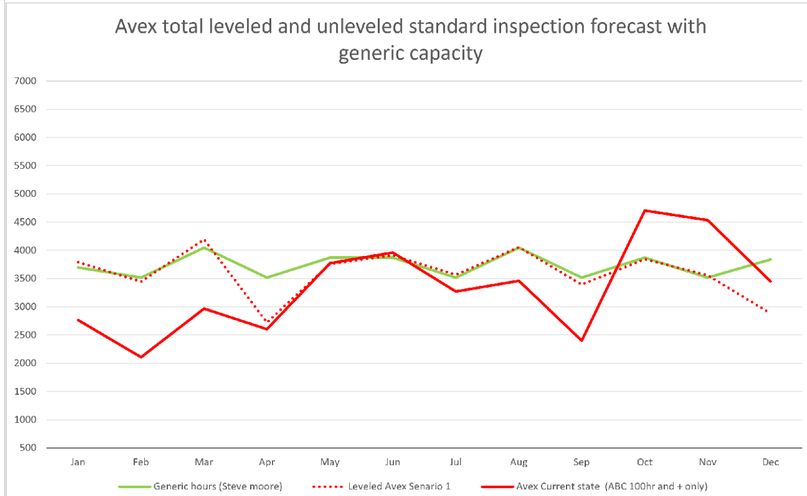
· Create a unified pitch for customers and the benefits to the customer if they participate in the leveling of the schedule.

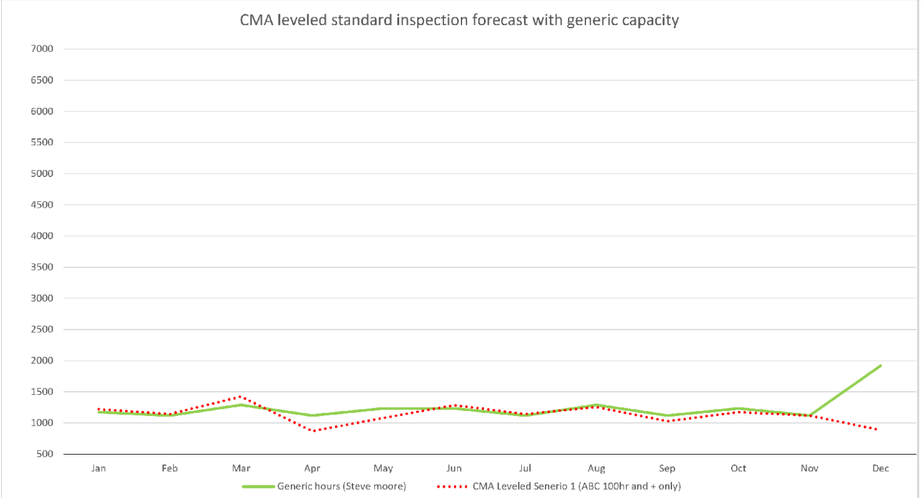
Define what incentives Avex is willing to use with customers to secure a leveled schedule.

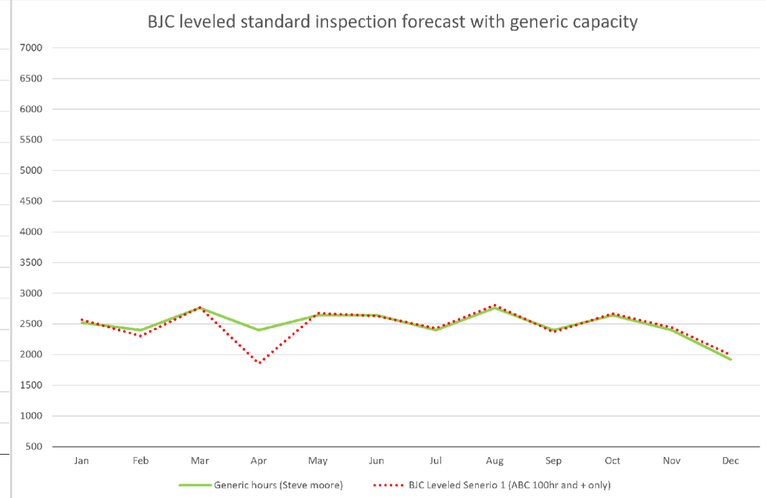
Quantify the cost of pilot services and aircraft use when switching customers between bases.

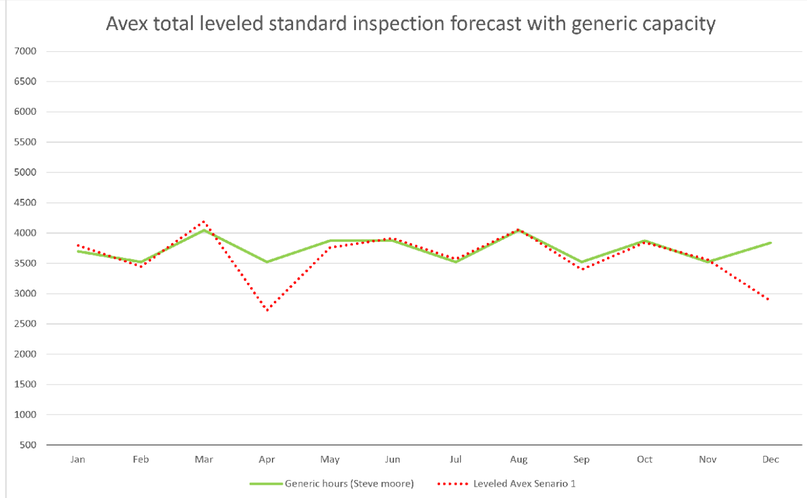

CMA leveled and unleveled standard inspection forecast with generic capacity
7000
6500
6000
5500
5000
4500
4000
3500
3000
2500
2000
1500
1000
500
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
Generic hours (Steve moore)
CMA Leveled Senerio 1 (ABC 100hr and + only)
CMA Current state (ABC 100hr and + only)

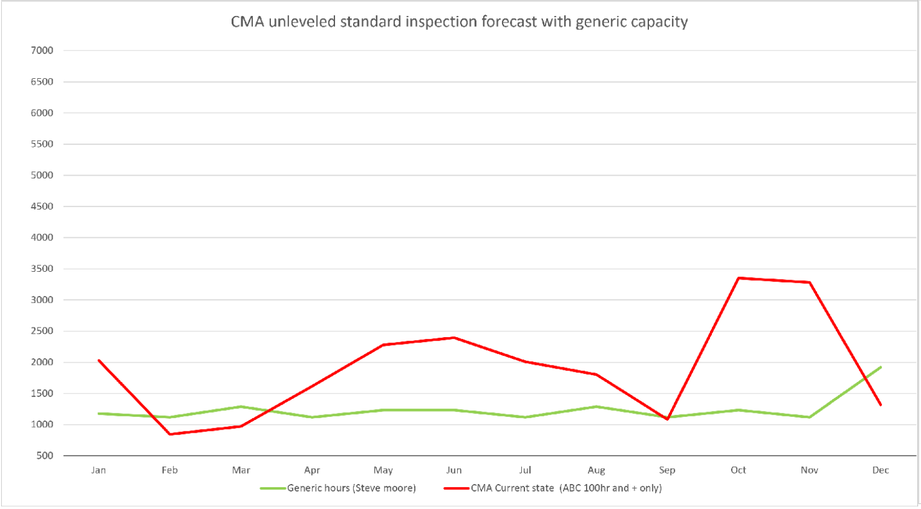

BJC leveled and unleveled standard inspection forecast with generic capacity
7000
6500
6000
5500
5000
4500
4000
3500
3000
2500
2000
1500
1000
500
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
Generic hours (Steve moore)
BJC Leveled Senerio 1 (ABC 100hr and + only)
BJC Current state (ABC 100hr and + only)

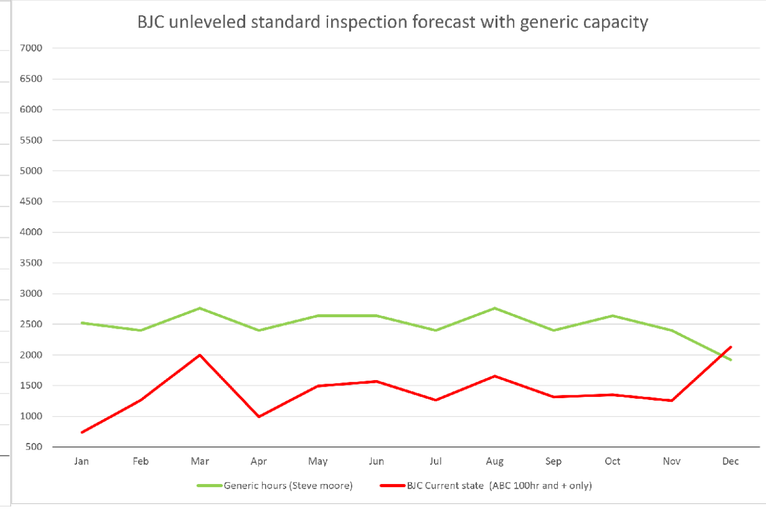

Avex total leveled and unleveled standard inspection forecast with generic capacity
7000
6500
6000
5500
5000
4500
4000
3500
3000
2500
2000
1500
1000
500
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
Generic hours (Steve moore)
Leveled Avex Senario 1
Avex Current state (ABC 100hr and + only)

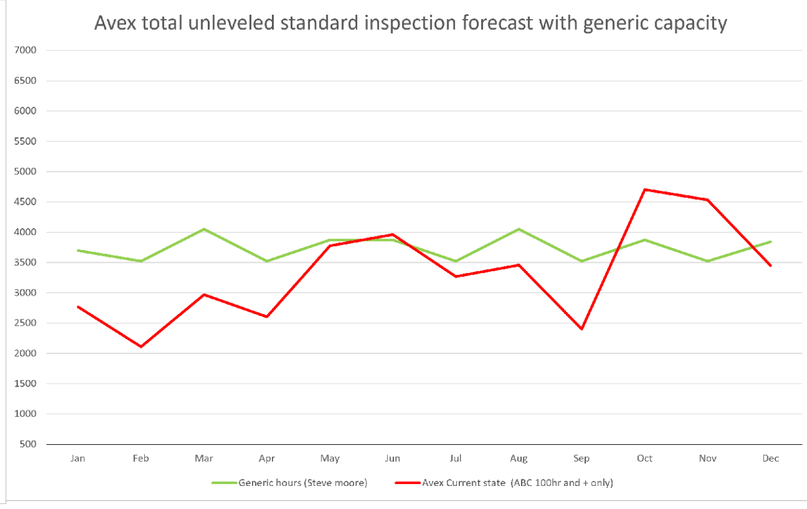

CMA leveled standard inspection forecast with generic capacity
7000
6500
6000
5500
5000
4500
4000
3500
3000
2500
2000
1500
1000
500
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
Generic hours (Steve moore)
CMA Leveled Senerio 1 (ABC 100hr and + only)


BJC leveled standard inspection forecast with generic capacity
7000
6500
6000
5500
5000
4500
4000
3500
3000
2500
2000
1500
1000
500
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
Generic hours (Steve moore)
BJC Leveled Senerio 1 (ABC 100hr and + only)


Avex total leveled standard inspection forecast with generic capacity
7000
6500
6000
5500
5000
4500
4000
3500
3000
2500
2000
1500
1000
500
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
Generic hours (Steve moore)
Leveled Avex Senario 1


CMA unleveled standard inspection forecast with generic capacity
7000
6500
6000
5500
5000
4500
4000
3500
3000
2500
2000
1500
1000
500
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
Generic hours (Steve moore)
CMA Current state (ABC 100hr and + only)


BJC unleveled standard inspection forecast with generic capacity
7000
6500
6000
5500
5000
4500
4000
3500
3000
2500
2000
1500
1000
500
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
Generic hours (Steve moore)
BJC Current state (ABC 100hr and + only)


Avex total unleveled standard inspection forecast with generic capacity
7000
6500
6000
5500
5000
4500
4000
3500
3000
2500
2000
1500
1000
500
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
Generic hours (Steve moore)
Avex Current state (ABC 100hr and + only)